
ECOLOGY, BIODIVERSITY & EVOLUTION-1(BT101)

CURRENT FINAL PAPER SPRING 2023(16August)

SHORT QUESTIONS(3)

1. What is nucleus made up of?

- 1) The nuclear envelope
- 2) Nucleoplasm and Chromatin (two major types of material)
- 3) The nucleolus
- 4) DNA

2. Height of stratospheres at equator and pole?

- 20 km (66,000 ft; 12 mi) at equator
- 10 km (33,000 ft; 6.2 mi) at midaltitude
- 7 km (23,000 ft; 4.3 mi) at the poles

3. Layers of atmosphere?

- 1) ☐ Troposphere: 0 to 12 km
- 2) ☐ Stratosphere: 12 to 50 km
- 3) ☐ Mesosphere: 50 to 80 km
- 4) ☐ Thermosphere: 80 to 700 km
- 5) ☐ Exosphere: 700 to 10,000 km

4. How many types of energy exist in ecosystem?

Energy exists in several different forms

- 1) Light
- 2) Chemical-bond energy
- 3) Motion
- 4) Heat

All organisms need energy to function

5. Stages of foraging Cycle.

- 1) Search
- 2) Assessment
- 3) Capture
- 4) Handling

6. Ozone depleting substance

- Halocarbon refrigerants
- Solvents
- Propellants in aerosols
- Chlorofluorocarbons
- Hydro chlorofluorocarbons
- Methyl bromide
- Carbon tetrachloride
- Methyl chloroform

7. Name the factors control of an ecosystem

- • **External**: Climate, topography
External factors are independent of internal factors.
- • **Internal**: Types of species, population size, disease, gene pool, reproduction rate
Internal factors are controlled by feedback loop

8. Panting?

Panting, a method of cooling, used by many mammals, most birds, and some reptiles, accomplished by means of the evaporation of water from respiratory surfaces, such as skin or mouth.

9. Define Taxis, Limiting factor & Range of Optimum?

Taxis: Directed movement of an organism in response to a stimulus.

Limiting factors: Environmental factors that restrict population growth or distribution.

Range of optimum: Specific range of conditions in which an organism thrives best, such as pH and temperature.

10. Differentiate between monogamous, polygamous and polyandrous?

Monogamous: A mating system where an individual has only one partner at a time.

Polygamous: A mating system where an individual has multiple partners of the opposite sex at the same time.

Polyandrous: A mating system where a female has multiple male partners at the same time.

11. What is aestivation?

Survival in extended period of drying. Avoid temperature damage and dehydration. Aestivation or “summer sleep”, on the other hand, is the low metabolic process by the animals during summers.

Examples: Insects, frog, turtle

12. What Is the Role of University of Wisconsin Madison Arboretum in Restoration ecology?

The University of Wisconsin-Madison Arboretum plays a significant role in restoration ecology. It has been a pioneer in the restoration of the prairie since November 1935. The arboretum's efforts have resulted in the transformation of the prairie into its current state.

13. Evidence of theory of natural selection?

Darwin based his theory on many solid evidences from his observations and collected samples. Three major evidences:

- 1) Geology
- 2) Fossil Evidences
- 3) Galápagos islands

14. Why theory of inheritance of acquired characteristics or soft inheritance rejected?

The inheritance of acquired characteristics is rejected because it violates the fundamental principles of genetics and evolution. Traits that are acquired during an organism's lifetime are not encoded in its DNA. Therefore, these traits cannot be passed down to its offspring. Inheritance is based on the genetic information that is passed down from parents to offspring through the DNA in gametes, such as sperm and eggs. This genetic information is not affected by environmental factors or changes that occur during an organism's lifetime. The theory of inheritance of acquired characteristics was proposed by Jean-Baptiste Lamarck in the early 19th century, but it was later disproven by the experiments of Gregor Mendel and the discovery of DNA.

15. Consumer & its types?

An organism that obtains energy and nutrients by consuming other organisms. Consumers are part of the food chain or food web and can be categorized into different trophic levels based on their feeding habits.

Types of Consumers

- 1) **Herbivore:** • Primary food source is plant-based. • Examples: vertebrates as well as invertebrates. • Digestive systems capable of handling large amounts of plant material. • There are eight types of herbivory depending upon different feeding habits.
- 2) **Carnivore:** • Primary food source is animal-based. • Examples: vertebrates as well as invertebrates • Digestive systems is more efficient in protein digestion.
- 3) **Omnivore:** An omnivore is an organism that eats plants and animals. The term stems from the Latin words *omnis*, meaning “all or everything,” and *vorare*, meaning “to devour or eat.” Example: humans.

16. Define Omnivores.

An omnivore is an organism that eats plants and animals. The term stems from the Latin words *omnis*, meaning “all or everything,” and *vorare*, meaning “to devour or eat.” **Example:** humans.

Types of Omnivores

- 1) **Frugivores:** Frugivores are animals that primarily feed on fruits. Examples include wolves and orangutans.
- 2) **Insectivores:** Insectivores are animals that primarily feed on insects and other small invertebrates. Swallows and pink fairy armadillos are examples of insectivores.
- 3) **Granivores:** Granivores are animals that specialize in consuming seeds and grains. Large ground finches and mice are examples of granivores.

17. Autotrophs, heterotroph, saprozoic ❖ Autotrophs

Organisms that can produce their own food using energy from the sun or inorganic substances.
Example: Plants perform photosynthesis to produce glucose.

❖ Heterotrophs

Organisms that obtain energy by consuming other organisms or organic matter. **Example:** Animals that obtain nutrients by eating plants or other animals.

❖ Saprozoic

A type of heterotrophic nutrition where organisms feed on decaying organic matter. **Example:** Fungi that obtain nutrients by decomposing dead plants or animals.

18. Mutualism

• Long-term relationship between individuals of different species • Both individuals get benefit
Example: An example of mutualism is the relationship between bees and flowers. Bees rely on flowers for nectar and pollen as a food source, while flowers rely on bees for pollination. As bees collect nectar from flowers, they transfer pollen from one flower to another, allowing the flowers to reproduce and produce seeds.

19. Difference between grazer and browsers?

❖ Grazers

• Feeds on plants such as grasses, soft flowering plants (forbs) and algae • Plants are not killed. •
Examples: Deer & Snails

❖ Browsers

• Feed on non-grass plant material • Leaves, soft shoots, twigs, fruits, woody plants such as shrubs
Examples: Goats, deer, giraffe, elephants

20. Diff btw mimicry and camouflage?

Camouflage

Camouflage, also known as cryptic coloration, is the use of any combination of materials, coloration, or illumination for the purpose of concealment from predators.

Example: Chameleons, which can change the color of their skin to blend in with their surroundings.

Mimicry

Mimicry is the resemblance of one species to another for protective or aggressive purposes. It is different from camouflage because instead of hiding through camouflage strategies, organisms that employ mimicry imitate the appearance, behavior, or other characteristics of a different species.

Example: The walking stick insect mimics the appearance of a twig or branch to blend into its surroundings.

21. Crypsis and change of skin color and counter shading

The ability of certain organisms to blend in with their surroundings by having colors and patterns that resemble their natural background.

Example: Egyptian nightjar, parrots

Changeable skin coloration:

- Actively change their skin patterns and colors.
- Change can occur rapidly or seasonally.
- Chameleons are known for their ability to rapidly change their coloration.
- Arctic foxes change their fur color from brown or grey in the summer to white in the winter.
- Animals use special chromatophore cells and fur molting to resemble their current background.

Countershading:

- Shadow makes the upper side darker and the underside lighter.
- Countershading involves graded coloration to counteract the effect of self-shadowing.
- It creates an illusion of flatness.
- Countershading makes the animal appear darker below and lighter above.
- **Example** animals: Gazelles, grasshoppers, sharks, dolphins.

22. Differentiate between autolysis and putrefaction

Autolysis and putrefaction are two processes that occur during the decomposition of dead tissue, but they differ in their underlying mechanisms and effects.

Autolysis:

The breakdown of tissues by the body's own internal chemicals and enzymes. Autolysis is the breakdown of cells and tissues by enzymes that are released from within the cells after death. This process is a natural part of cellular metabolism and occurs in all living cells, but it is accelerated after death.

Putrefaction:

Putrefaction, on the other hand, is the breakdown of tissues by bacteria and other microorganisms that colonize the body after death. This process leads to the release of gases and foul odors, as well as the breakdown of tissues into a liquid or semi-liquid state.

The breakdown of tissues by bacteria, resulting in the release of compounds like cadaverine and putrescine, which produce a decaying odor. Proteolysis releases nitrogen and phosphorus.

- Carbohydrate breakdown:
 - Aerobic breakdown by fungi and bacteria.
 - Anaerobic breakdown by bacteria.
- Anaerobic degradation of lipids.
- Degradation of nucleic acids and bones.

23. Note on R-Selected Species?

Traits of r-selected species

- R-selected species prioritize quantity over quality in reproduction.
- R-selected species have a high growth rate.
- R-selected species produce a large number of offspring.
- R-selected species have a relatively low chance of survival to adulthood.
- R-selected species have high reproductive rates.
- R-selected species have small body sizes.
- R-selected species reach maturity early.
- R-selected species have short generation times.
- R-selected species disperse their offspring widely.
- Examples of r-selected species include bacteria, grasses, cephalopods, and rodents.

24. Ecological Pyramids?

An **ecological pyramid** is a graphical representation of the relationship between different organisms in an ecosystem. It shows the number, biomass, and energy of different trophic levels in an ecosystem.

There are three different **types of ecological pyramids** to illustrate ecosystems:

- 1) **Pyramid of energy:** represents how much energy is available in each trophic level
- 2) **Pyramid of numbers:** represents the actual number of organisms present in each trophic level
- 3) **Pyramid of biomass:** represents the total mass of living things in each trophic level

25. Ecological niche ?

In a community, each organism faces the challenge of survival in its own unique way. It occupies a specific functional role and place called its niche. An organism's niche is defined by how it uses the resources available in its environment. This includes how it utilizes space, consumes food, tolerates temperature ranges, finds suitable conditions for mating, and has specific moisture requirements.

The niche of an organism can be described by looking at these factors:

- how it occupies space,
- what it eats,
- its temperature preferences,
- its mating requirements,
- and its need for moisture.

26. Top-down & Bottom-up Effect?

❖ Top-down effect

Top-down effects occur when the presence or behavior of top predators or higher trophic levels impact the lower trophic levels and the overall ecosystem structure.

❖ Bottom-up effects

Bottom-up effects occur when changes in the availability of resources or primary producers, such as plants or algae, influence the higher trophic levels.

27. Types of Genetic Drift?

Founder Effect: Limited genetic diversity in a new population established by a small group of individuals.

Bottleneck Effect: Significant loss of genetic variation due to a drastic reduction in population size.

28. Explain parapatric speciation with example.

Parapatric speciation occurs when two populations of a species evolve into separate species while still sharing a common border or overlapping range.

An **example** of parapatric speciation is the evolution of different species of butterflies within a mountain range. As the altitude changes, different populations of butterflies adapt to the specific conditions at each elevation. Over time, these populations become reproductively isolated and develop distinct traits, leading to the formation of separate butterfly species within the same mountain range.

29. justify how the evolution of fourteen different species of Darwin finches on the Galapagos Island related to the allopatric speciation?

The evolution of fourteen different species of Darwin's finches on the Galapagos Islands is a classical example of allopatric speciation. Populations became isolated on various islands, leading to adaptations to new habitats and food resources. Mutations increased variations, and natural selection favored traits that promoted successful reproduction. The combined forces of isolation, mutation, and natural selection allowed the finches to diverge into specialized species with unique feeding habits.

30. How pollution is classified ?

Pollution Classified as:

- 1) Point source
- 2) Non point source

Point source: Single identifiable source of pollution. *Example:* • Land fill • Factory drain • Sewages pipes

Non Point source: Resulting from many diffuse sources. *Example:* • Sediment • Nutrients • Toxic contaminants and chemicals • Pathogens

31. What is cloud pollution?

Cloud pollution, also known as cloud computing pollution, refers to the environmental impact caused by the energy consumption and carbon emissions associated with data centers and cloud computing infrastructure.

32. Light pollution?

Light pollution is a type of pollution that occurs when artificial light is overused or misused, leading to excessive brightness and glare in the night sky. This can have a number of negative effects, such as disrupting natural habitats and ecosystems, interfering with astronomical observations, and disrupting human sleep patterns and circadian rhythms. Light pollution can also waste energy and contribute to climate change. To help reduce light pollution, it is important to use lighting fixtures that are designed to minimize glare and direct light downward, and to turn off lights when they are not needed.

33. Write the sources of wastewater.

- 1) Domestic wastewater from households
- 2) Municipal wastewater from communities (also called sewage)
- 3) Industrial wastewater from industrial activities
- 4) Agricultural wastewater from agricultural activities
- 5) Storm water

34. Important phases/stages of the water cycle?

Here are the names of important parts of the water cycle:

- 1) Evaporation
- 2) Condensation
- 3) Precipitation
- 4) Surface Runoff
- 5) Percolation(Infiltration)
- 6) Groundwater flow
- 7) Transpiration
- 8) Sublimation

35. Evaporation, percolation, ground water, transpiration?

Evaporation

Evaporation is the process by which a liquid (such as water) changes into a gas or vapor, usually due to an increase in temperature or a decrease in pressure. During evaporation, the molecules of the

liquid gain enough energy to break free from the surface and become a gas. This process is an important part of the water cycle, as it contributes to the formation of clouds and precipitation, which are important for the distribution of water on Earth.

Percolation (Infiltration)

Percolation is an important process. Rain water soaks into (infiltrates) the ground, into the soil and underlying rock layers. Some of this water ultimately returns to the surface at springs or in low spots downhill. Some of the water percolates underground It is called groundwater.

Groundwater

Water moves through the soil and rock layers. Many of the impurities in the water are filtered out. This filtering process helps clean the water.

Transpiration

Plants absorb water from the soil. Water moves from the roots through the stems to the leaves. Water reaches the leaves. Some of it evaporates from the leaves, adding to the amount of water vapor in the air. This process of evaporation through plant leaves is called transpiration. In large forests, an enormous amount of water will transpire through leaves.

36. Benefits of atmosphere

- Atmosphere is responsible for sustaining life on earth.
- Without it, earth will be barren like moon and mars.
- Atmosphere is key element in survival of life.

The Earth's atmosphere protects and sustains the planet's inhabitants by providing warmth and absorbing harmful solar rays. In addition to containing the oxygen and carbon dioxide, which living things need to survive, the atmosphere traps the sun's energy and wards off many of the dangers of space.

37. Stratosphere?

The stratosphere is the layer of the Earth's atmosphere located above the troposphere and below the mesosphere. It extends from about 12 to 50 kilometers (6 to 31 miles) above the Earth's surface. The stratosphere contains the ozone layer, which absorbs much of the sun's ultraviolet radiation. This layer helps protect life on Earth from the harmful effects of excessive UV radiation. The stratosphere is also characterized by a temperature inversion, where temperatures increase with altitude due to the absorption of solar energy by ozone molecules.

38. Explain evolution of 3rd layer(mesosphere) of atmosphere?

The third atmosphere of our planet witnessed a constant rearrangement of continental plates, leading to ever-changing landmasses. This dynamic process contributed to the inflow and outflow of carbon dioxide (CO₂) from large continental carbonate stores, impacting the composition of the atmosphere. During this time, oxygen (O₂) synthesized through photosynthesis played a crucial role in reducing materials, but as time progressed, a shift occurred towards an oxidizing atmosphere. Approximately 541 million years ago, a significant milestone was reached as O₂ levels stabilized at around 15% of the atmosphere, setting the stage for further evolutionary developments on Earth.

39. Write down the properties thermosphere.

The upper layers of the atmosphere, specifically the thermosphere and exosphere, experience high temperatures due to absorption of intense solar radiation. Temperatures can reach extremely high levels, but they drop significantly when solar activity is limited. This region has a very low density, and molecules are too far apart to transfer heat through conduction. Sound cannot travel because the density is too low for molecular interactions. The area is influenced by solar winds, extreme UV storms, and atmospheric waves. The bombardment of particles is intense, which prevents the weakening of the magnetic field. The boundary between the thermosphere and exosphere is difficult to distinguish.

The exosphere is the thin, outermost layer surrounding a planet, where molecules are gravitationally bound but have such low density that they don't behave like a gas. Eventually, the atmosphere thins out and merges with interplanetary space. Our understanding of this region is limited due to a lack of research.

40. Cytochrome c?

• Cytochrome c is a protein present in the cellular respiration pathways in all eukaryotic organisms. • Closely related organisms have similar cytochrome c molecules. • Structure of cytochrome c has changed so little during hundreds of millions of years. • It is said to have been conserved evolutionarily. • Not all proteins are conserved as rigorously as cytochrome c.

41. Fitness?

Adaptedness or fitness is a measure of the capacity for successful reproduction in a given environment. Not every characteristic is an adaptation to some kind of environmental situation. Neither adaptations lead to perfection.

- Fitness is quantified as reproductive success
- The number of surviving offspring left in the next generation.
- The most fit phenotype is simply the one that produces, on average, the greatest number of offspring.

42. Frequency dependent Selection?

This type of selection favors certain phenotypes depending on how commonly or uncommonly they occur. • It has two types:

- 1) Negative
- 2) Positive

43. Difference between oscillating and Frequency dependent selection?

❖ Oscillating selection

• In oscillating selection, the fitness of a phenotype does not depend on its frequency. • Rather, environmental changes lead to the oscillation in selection.

❖ Frequency dependent selection

• In frequency dependent selection, it is the change in frequencies themselves. • Fitness of the different phenotypes can vary according to their frequencies and environment.

44. Polymorphism?

Occurrence of distinct variations within a population, without intermediate or overlapping phenotypes. It highlights the genetic diversity and variability within a species.

Example: An example of polymorphism is the color variation in butterfly populations. Within a single species, there may be individuals with different wing colors or patterns, such as black, orange, or a combination of both. These distinct forms coexist in the population without intermediate colors or patterns. This polymorphism can be advantageous for the butterflies, as it helps them blend in with their environment and avoid predators.

45. Types of Flagella?

- 1) **Monotrichous**; one flagella
- 2) **Lophotrichous**; tuft at one end
- 3) **Amphitrichous**; both ends
- 4) **Peritrichous**; all around bacteria.

46. Classification of Class Reptilia?

- 1) **Order Testudines**: Tortoises, turtles
- 2) **Order Squamata**: Snakes, lizards, worm lizards, Lizard like reptiles, single species, Sphenodon and Tuataras
- 3) **Order Crocodilia**: Crocodiles, alligators, gavials

47. Classes of Phylum Arthropoda?

- 1) **Insecta** (Insects): Examples include butterflies, ants, bees, and beetles.
- 2) **Arachnida** : Examples include spiders, scorpions, ticks, and mites.
- 3) **Crustacea** : Examples include crabs, lobsters, shrimp, and barnacles.
- 4) **Diplopoda**: Examples include millipedes.
- 5) **Chilopoda** : Examples include centipedes.
- 6) **Merostomata** : Examples include horseshoe crabs.

48. Economic importance of Phylum Myxozoa?

• The most economically important myxosporean is *Myxosoma cerebralis*. • Which infects the nervous system and auditory organs of trout and salmon. • Causing whirling or tumbling disease

49. Zoomastigophora?

• Lack chloroplast • Heterotrophic or saprozoic • Parasites of humans

Examples: 1. *Trypanosoma brucei* • It has three sub species.

- 1) **T. b. brucei**: Parasite of non human • Common host are mammals of Africa
- 2) **T. b. gambiense**: • Causes sleeping sickness in humans
- 3) **T. b. rhodesiense**: • Causes sleeping sickness

50. Which diseases are caused by apicomplexan?

Here are some diseases caused by apicomplexan parasites

- 1) **Malaria** - caused by the parasite *Plasmodium*.
- 2) **Toxoplasmosis** - caused by the parasite *Toxoplasma gondii*.
- 3) **Babesiosis** - caused by various species of the parasite *Babesia*.
- 4) **Cryptosporidiosis** - caused by the parasite *Cryptosporidium*.
- 5) **Coccidiosis** - caused by the parasite *Eimeria*.

51. Characteristics of Nematodes?

From handouts

• Triploblastic • Bilateral • Vermiform • Unsegmented • Pseudocoelomate • Complete digestive tract • Unique excretory system comprised of two or more renette cell • Body wall has only longitudinal muscles • Cuticle may contain spines, bristle papillae and ridges • Pseudocoelom is a spacious fluid filled cavity • Visceral organs • Form hydrostatic skeleton • May be herbivores, carnivores, omnivores and saprotrophs • Feed on blood and tissue fluid of their hosts • Complete digestive tract • Excretion of nitrogenous waste • Ventral gland cells, called renette • Nervous system consist of anterior nerve ring

From internet

- Nematodes are roundworms found in various habitats.
- They have a cylindrical body and complete digestive system.
- Nematodes can be free-living or parasitic.
- They reproduce sexually and play important roles in ecosystems.
- There are millions of nematode species, with many yet to be identified.
- Nematodes have a protective outer cuticle.
- They have a simple nervous system.
- Some nematodes cause diseases in plants, animals, and humans.
- They can be found in soil, freshwater, and marine environments.
- Nematodes are incredibly abundant in nature.
- Nematodes have a bilateral symmetry.
- They can range in size from microscopic to several meters long.
- Some nematodes have the ability to survive extreme conditions, such as freezing temperatures or drought.

52. Circulatory system of Nematodes?

Nematodes have a relatively simple circulatory system. They lack a true circulatory system with specialized blood vessels, but instead rely on a fluid-filled body cavity called the pseudocoelom to transport nutrients and gases. This fluid, called pseudocoelomic fluid, helps distribute nutrients and oxygen throughout the body and aids in waste removal. However, nematodes do not have a dedicated organ for pumping and circulating this fluid like a heart.

53. Reproductive System of Arthropods

In terms of reproduction, most arthropods are dioecious and have paired genital openings. Males often package sperm in a spermatophore and transfer them into the female genital tract during copulation. Sperms are transferred via modified pedipalps.

54. Coxal Glands & Malpighian Tubules?

Arthropods have two main types of excretory organs:

- coxal glands and
- Malpighian tubules.

1. Coxal Glands:

Coxal glands are paired structures with thin walls that absorb nitrogenous waste and excrete them through an excretory pore.

2. Malpighian Tubules:

Malpighian tubules, on the other hand, absorb waste material from the blood and empty it into the gut. The major excretory product of Malpighian tubules is uric acid, which is excreted as a semisolid with little water loss.

55. Explain the Characteristics of Echinoderms?

• 7000 species are exclusively marine • Pentaradial symmetry • Skeleton consists of a series of calcium carbonate plates called Ossicles • Water vascular system • Series of water-filled canals, and their extensions called tube feet • Hemal system derived from coelomic cavities • Nervous system consist of nerve net, nerve ring and radial nerves

56. How echinoderms regenerate?

Echinoderms, such as starfish, have the incredible ability to regenerate their entire body from just a portion of the central disk. If a starfish's arm containing a portion of the central disk is broken off, it can regenerate a whole new organism from that fragment. This process of regeneration is facilitated by the starfish's unique cellular properties and genetic mechanisms.

57. Function of Water vascular system in Echinodermata?

The water vascular system in echinoderms, such as sea stars and sea urchins, serves multiple functions. It helps with locomotion, respiration, and feeding. The system consists of a network of fluid-filled canals and tube feet. The tube feet are used for movement, attachment, and capturing

prey. Water is circulated through the system by muscular contractions, allowing for coordinated movement and sensory perception.

58. Characteristics of chordates

- Chordates have a notochord, a flexible rod-like structure that provides support.
- They possess a dorsal nerve cord, which develops into the spinal cord in vertebrates.
- Chordates have pharyngeal slits or clefts, which may function as respiratory structures or filter-feeding mechanisms.
- They have a post-anal tail that extends beyond the anus.
- Chordates have an endostyle or thyroid gland, which plays a role in metabolism and hormone regulation.
- Chordates are a diverse group of animals.
- They have a bilateral symmetry.
- Chordates have a coelom, a body cavity that contains their internal organs.
- They possess a closed circulatory system.
- Chordates have a well-developed nervous system.
- Chordates exhibit segmentation, with their bodies divided into repeating units.
- They have a complete digestive system with a mouth and an anus.
- Chordates possess a well-developed muscular system for movement.
- They have a high level of cephalization, with a concentration of sensory organs and a distinct head region.
- Chordates exhibit a wide range of reproductive strategies, including internal fertilization and external egg-laying.

59. Characteristics of Pisces?

The class Pisces refers to the group of animals known as fish. Some characteristics of this class include:

- Fins and tail for locomotion
- Can be very fast swimmers like sharks
- Endoskeleton of cartilage or bones
- Use gills for respiration
- - Fish are cold-blooded vertebrates that live in water.
- - They have gills that allow them to extract oxygen from the water.
- - Most fish have scales that protect their bodies.
- - Fish have fins that help them swim and maneuver in the water.
- - They have a streamlined body shape that reduces drag in the water.
- - Fish use a variety of methods to reproduce, including external fertilization and internal fertilization.
- - They have a lateral line system that helps them detect changes in water pressure and vibrations.
- - Fish have a variety of feeding habits, including herbivorous, carnivorous, and omnivorous.
- - They exhibit a wide range of sizes, from tiny species like the pygmy goby to large species like the whale shark.
- - Fish have adapted to various aquatic environments, including freshwater, saltwater, and brackish water.

60. Digestion in pisces?

- Herbivore, carnivore and omnivore
- Complete gut
- With stomach/without stomach
- Liver, pancreas, pyloric caeca and endocrine system help in digestion.

In fishes (Pisces), digestion involves various adaptations and organs that aid in the breakdown and absorption of nutrients. Fishes can have different feeding habits, including herbivores, carnivores, and omnivores. They possess a complete gut, consisting of a mouth, esophagus, stomach, and intestines. While most fishes have a stomach, its structure and function can vary. The liver produces bile to aid in the breakdown of fats, while the pancreas produces enzymes for carbohydrate, protein, and fat digestion. Pyloric caeca, finger-like projections in the intestines, increase the surface area for nutrient absorption. The endocrine system regulates digestion. Altogether, these adaptations and organs work together to support efficient digestion in fishes.

61. Characteristics of Mammals?

- - Mammals are warm-blooded vertebrates.
- - Mammals have hair or fur on their bodies.
- - Mammals have mammary glands that produce milk for feeding their young.
- - Mammals have a specialized jaw structure with different types of teeth.
- - Mammals have a four-chambered heart.
- - Mammals have a diaphragm for breathing.

- - Mammals give birth to live young (except monotremes like platypus and echidnas).
- - Mammals have a highly developed brain and complex behavior.
- - Mammals have a unique middle ear structure with three bones (malleus, incus, and stapes).
- - Mammals have a neocortex, which is responsible for higher cognitive functions.
- - Mammals have a varied range of body sizes, from the tiny bumblebee bat to the massive blue whale.
- - Mammals exhibit diverse feeding habits, including herbivory, carnivory, and omnivory.
- - Mammals have a well-developed respiratory system, with lungs for breathing air.
- - Mammals have a highly efficient circulatory system, with a closed network of blood vessels.
- - Mammals have a complex social structure and exhibit parental care towards their offspring.

62. What are conserved structures in birds?

One of the conserved structures in birds is their skeletal system, which includes features such as the beak, wings, and specialized bones for flight. Another conserved structure is their feathers, which play a crucial role in flight, insulation, and display. Additionally, birds have a unique respiratory system, including air sacs and lungs, that enables efficient oxygen exchange during flight.

63. What is endomembrane system?

The endomembrane system is a series of compartments that work together to package, label, and ship proteins and molecules. In your cells, the endomembrane system is made up of both the endoplasmic reticulum and the Golgi apparatus. These compartments are folds of membranes that form tubes and sacs in your cells.

64. Composition of mitochondrial membrane

The mitochondrial membrane is made up of the outer mitochondrial membrane (OMM) and the inner mitochondrial membrane (IMM). The OMM is porous and involved in molecule transport, while the IMM is highly folded and responsible for energy production through oxidative phosphorylation.

65. Functions of Peroxisomes?

Peroxisomes have several functions, including:

- - Breaking down fatty acids and amino acids through oxidation
- - Detoxifying harmful substances, such as hydrogen peroxide
- - Participating in the synthesis of certain lipids
- - Facilitating photorespiration (glycolate pathway)
- - Contributing to the metabolism of certain molecules, like purines
- - Assisting in the breakdown of long fatty acids into shorter chains, which are then sent to mitochondria for further breakdown

66. Define microfilament and it's funtion?

Microfilaments, also known as actin filaments, are thin structures composed of actin proteins that form a 3-D network inside the cell membrane. They have multiple **functions**, including facilitating cell movement, muscle contraction, intracellular transport, maintaining cell shape, aiding in cytokinesis, and enabling cytoplasmic streaming. They can form networks and bundles within the cell, allowing for the movement of organelles and the generation of mechanical forces. Overall, microfilaments are essential for the proper functioning and organization of cells.

67. Endoplasmic Reticulum

The endoplasmic reticulum (ER) is an organelle found in eukaryotic cells. It plays a crucial role in protein synthesis, lipid metabolism, and calcium storage. The ER consists of two main regions: the rough ER, which is studded with ribosomes and involved in protein synthesis, and the smooth ER, which lacks ribosomes and is involved in lipid synthesis and detoxification. The ER is interconnected with the nuclear envelope and other organelles, allowing for the transport of molecules within the cell.

68. Characteristics of bacteria

- **Prokaryotic:** Bacteria are prokaryotic organisms, lacking a nucleus and membrane-bound organelles.
- **Cell Wall:** Bacteria have a cell wall made of peptidoglycan, providing structure and protection.
- **Reproduction:** Bacteria reproduce through binary fission, dividing into two identical daughter cells.
- **Metabolic Diversity:** Bacteria exhibit a wide range of metabolic capabilities, including autotrophic and heterotrophic modes of nutrition.
- **Ubiquitous:** Bacteria can be found in diverse environments, playing important roles in ecological processes.
- Bacteria can exchange genetic material and adapt to different environments.

LONG QUESTIONS(5)

1. Analogy and homology

Homology involves the study of organs that have evolved from same origin or from common ancestor but have different functions. **For example**, wings of bat and hands of humans.

Analogy involves the study of organs that are evolved from different ancestors but perform a similar function. **For example**, wings of bat and wings of insects, which have similar function but different structure.

2. Types of Grazing ?

❖ Graminivory

- Feeding primarily on grass • Specifically "true" grasses in family Poaceae

Examples: Horses, cattle, hippopotamus, grasshoppers

❖ Coprophagy

- Feeding on grass, forbs, leafy weeds, fruits, tree bark • Heavy grazing for half an hour of grazing period • Followed by half an hour of selective feeding. Later feeding in intervals • Eat their or other species's soft feces (caecotrophs)

Example: Rabbit

3. Differentiate Monocot & Dicot.

➤ Monocots:

- Have one cotyledon
- Have parallel veins in leaves
- Have scattered vascular tissue
- Have flower parts in multiples of three
- Have seeds with one embryonic leaf
- Have fibrous roots
- **Examples** include grasses, lilies, and orchids

➤ Dicots:

- Have two cotyledons
- Have branching veins in leaves
- Have circular arrangement of vascular tissue
- Have flower parts in multiples of four or five
- Have seeds with two embryonic leaves
- Have taproots
- **Examples** include roses, beans, and oak leave

4. Origin of Eukaryotic Cells

Endosymbiotic Theory

- Chloroplasts and Mitochondria evolved from symbiotic bacteria that lived inside of a larger prokaryote. • Also known as Symbiogenesis • Chloroplasts evolved from endocytosis of cyanobacteria
- Mitochondria evolved from endocytosis of aerobic bacteria.

The endosymbiotic theory, which suggests that eukaryotic cells evolved from a symbiotic relationship between different types of prokaryotic cells. According to this theory, a host cell engulfed a smaller prokaryotic cell, forming a mutually beneficial relationship. Over time, the engulfed cell evolved into organelles such as mitochondria and chloroplasts, which are key features of eukaryotic cells. This theory explains the presence of double membranes and similarities in genetic material between these organelles and prokaryotes.

5. Characteristics of Mastigophora?

- Unicellular or colonial
- Locomotion by flagella, pseudopodia, or both
- Autotrophic (self-nourishing), saprozoic (living in decaying organic matter), or heterotrophic (obtains energy from organic compounds)
- Single type of nucleus
- Sexual reproduction (usually)

Phylum Mastigophora has **two classes**.

- 1) Phytomastigophora
- 2) Zoomastigophora

6. Detail note on Different Types of Pseudopodia?

When feeding and moving, amoeba forms temporary cell extensions called **pseudopodia**.

Types of pseudopodia

Majorly amoeba has four types

- 1) Lobopodia

- 2) Filopodia
- 3) Reticulopodia
- 4) Axopodia

❖ **Lobopodia**

- Contain ectoplasm and endoplasm
- Used for locomotion
- Used for engulfing feed

❖ **Filopodia**

- Only ectoplasm in pseudopodia
- Two-way streaming
- Provides food in conveyor belt fashion

❖ **Reticulopodia**

- Similar to filopodia
- Branch and rejoin to form net like series of cell extensions

❖ **Axopodia**

- These are thin filamentous supported by central axis of microtubules.
- Food caught on axopodia delivered to the central cytoplasm of amoeba.

7. Differentiate between methanogen and halophiles?

❖ **Methanogens**

- They release methane (CH₄) as a waste product
- Many live in mud at the bottom of lakes and swamps because it lacks oxygen
- Some live in the intestinal tracts of animals to help break down food.
- They could play a role in garbage/sewage cleanup by eating garbage.

❖ **Halophiles**

- Halo = salt
- phil = loving
- Can live in water with salt concentrations exceeding 15%.
- The ocean's concentration is roughly 4%.
- One habitat of halophils is Great Salt Lake.
- Great Salt Lake is actually three to five times saltier than the ocean.
- 2.5 million tons of sodium chloride is extracted from the lake.

8. Mitochondria

- Mito – thread
- Chondrion – granule like

Size and Shape:

- Mitochondria range in size from 0.05 to 1.0 µm.
- They can be 1 to 10 µm long.
- They are typically bean-shaped, but in fibroblast cells, they appear elongated and thread-like.

Number:

- The number of mitochondria in a cell depends on its type, size, and functional state.
- An average liver cell contains around 1500 mitochondria.

Structure:

- Mitochondria have a double membrane structure.
- The outer membrane and inner membrane enclose different compartments.
- The inner membrane is folded into structures called cristae.
- The space between the two membranes is called the intermembrane space.
- The innermost part is the matrix, which contains mitochondrial DNA and enzymes.

Function:

Mitochondria are crucial organelles involved in aerobic respiration, producing ATP, synthesizing mtDNA, RNA, and proteins, and storing calcium ions. They also participate in oxidation reactions, metabolic processes, and energy production through oxidative phosphorylation, the Krebs' Cycle, beta oxidation, and ammonia detoxification. Their structure includes an outer membrane, inner membrane with folded cristae, and a matrix containing DNA and enzymes. Overall, mitochondria play a vital role in cellular function and survival.

9. Structure & Function of Golgi Apparatus?

Golgi Body:

- Named after Camillo Golgi

- An Italian biologist who discovered the organelle with a light microscope in 1897.

Structure :

- Located near the end of the ER close to the nucleus
- Composed of several layers of cisternae (fluid-filled sacs)

Has three different parts:

- 1) cis-Golgi
- 2) medial-Golgi
- 3) trans-Golgi

Functions :

- - The cis side of the Golgi body is the first cisternae, while the trans side is the final cisternae.
- - Proteins are packaged into vesicles from the trans side, destined for lysosomes, secretory vesicles, or the cell surface.
- - Transport vesicles shuttle between the ER and Golgi bodies.
- - The cis face of the Golgi body receives transport vesicles from the ER, while the trans face releases vesicles.
- - The Golgi body marks and sorts molecules into different groups.
- - These groups are sent in secretory vesicles either inside or outside of the cell.
- - Proteins are modified in the Golgi body by adding specific sugar molecules to a core oligosaccharide attached to the protein.

10. Explain Characteristics & Classification of Platyhelminthes?

❖ Characteristics

- Usually flattened dorsoventrally
- Triploblastic
- acoelomate
- Bilaterally symmetrical
- Unsegmented worms
- Beginning of cephalization with an anterior cerebral ganglion
- Longitudinal nerve cords
- Protonephridia as excretory structures
- Hermaphroditic
- Complex reproductive systems

❖ Classification

- 1) Class Turbellaria (Free-Living Flatworms)
- 2) Class Monogenea
- 3) Class Trematoda (Flukes)
- 4) Class Cestoidea (Tapeworms)

11. Explain reproduction in arthropods.

Arthropods reproduce by sexual reproduction, which involves the generation and fusion of gametes. Most arthropods are either male or female, and they undergo internal fertilization. Once the egg has been fertilized, the female usually lays the egg, and it continues developing outside of the mother's body. However, some arthropods are hermaphroditic, meaning they have the organs of both sexes, such as in barnacles. Few crustaceans and insects can reproduce by means of parthenogenesis. They revert to reproducing sexually when conditions become more favorable again.

12. Digestive System of Echinodermata?

- Complete digestive tract that may be secondarily reduce
- Mouth open to a short esophagus
- Large stomach
- Stomach is divided into two region
- Larger, oral stomach (cardiac stomach)
- Smaller, aboral stomach (pyloric stomach)
- Absorptive structure called pyloric caecae
-

13. Metamorphosis & its types?

Metamorphosis is a biological process in which an organism undergoes a profound change in its body structure and function. It involves a series of distinct stages, during which the organism's body undergoes significant transformations. This process is commonly observed in insects, amphibians,

and some other animals, enabling them to adapt to different ecological niches and fulfill various roles in their life cycles.

TYPES

There are two main types of metamorphosis:

- **Incomplete Metamorphosis:**

This type of metamorphosis has three stages: egg, nymph, and adult. Insects like grasshoppers, crickets, and cockroaches undergo incomplete metamorphosis. The nymphs resemble miniature versions of the adults and gradually develop into adults through a series of molts.

- **Complete Metamorphosis:**

This type of metamorphosis has four stages: egg, larva, pupa, and adult. Insects like butterflies, beetles, and flies undergo complete metamorphosis. The larval stage, also known as a caterpillar or grub, looks completely different from the adult and has its own specialized feeding habits. The larva then enters the pupal stage, during which it undergoes a complete transformation inside a protective cocoon or pupal case. Finally, the adult emerges from the pupa with fully developed.

14. Binomial Nomenclature

- The binomial system of nomenclature brings order to a chaotic world of common names.
- Common names vary from country to country, and from region to region within a country.
- There was need of globally recognized names.
- In binomial system, no two kinds of animals have the same binomial name.
- Every scientific name has two parts:

- 1) First from genus
- 2) Second from species

- Names are written in italics

Example:

- Common name: Indian frog
- Genus name: *Rana*
- Species name: *tigrina*
- Scientific name: *Rana tigrina*

- This scientific name is universally accepted.

15. Theory of inheritance and soft inheritance?

The theory of "inheritance of acquired characteristics" or "soft inheritance" is the idea that traits that are acquired by an organism during its lifetime can be passed on to its offspring. This theory was proposed by Jean-Baptiste Lamarck in the early 19th century. However, this theory has been largely discredited by modern genetics and evolutionary theory, which suggest that genetic information is passed down through DNA and that acquired characteristics are not inherited.

- If an organism developed a characteristic feature through adapting to a new way of life during its lifetime, it would pass this on to its offspring
- The classic example given is that of the giraffe's neck.
- As the giraffe's ancestors searched for a richer food supply they stretched to reach higher branches in trees.
- Thus their stretched bodies were passed onto their offspring.

16. Explain genetic drift with example.

Genetic drift is the random change in allele frequencies in a population over time

Genetic drift is the random changes to the genetic makeup of populations living in the patches. It also leads to reduction in genetic diversity and increases homozygosity. Overall higher population sizes are favored in natural selection as smaller populations are weak in adaptations and are susceptible to changes. As a result smaller populations are at the risk of elimination. Habitat fragmentation also affects the predator-prey relationship as the division of habitat affects the population dynamics either by reducing the available space for the prey to hide from predators or by making it hard for predator to find its prey. Either way one of the two will be at disadvantage.

One **example** is the bottleneck effect. Imagine a population of elephants that face a severe natural disaster, resulting in only a few individuals surviving. The surviving elephants may have a different set of alleles compared to the original population. As the population recovers, the allele frequencies may be different from before the disaster due to the random chance of which alleles were passed on. This is an example of genetic drift, as the change in allele frequencies is not driven by natural selection but by chance events.

17. How Evolution & Paleontology related to each other? Diff b/w Evolution & Paleontology.

Paleontology

Paleontology is the study of fossils, which are the preserved remains or traces of ancient organisms. Paleontologists use fossils to reconstruct the history of life on Earth and to understand how organisms

have changed over time. Paleontologists use a variety of techniques to study fossils, including carbon dating, radiometric dating, and comparative anatomy.

Evolution

Evolution is the process by which species change over time through the process of natural selection. It is supported by a wide range of evidence from many different fields of biology, including genetics, ecology, and comparative anatomy.

Evolution and paleontology are closely intertwined. Paleontology, the study of fossils, provides valuable evidence that supports the theory of evolution. Fossils serve as a record of past life forms, allowing scientists to trace the evolution of species over time. By examining fossils found in different rock layers, paleontologists can uncover important information about ancient ecosystems, the timing of evolutionary events, and the relationships between different species. This evidence helps us understand the process of evolution and how species have changed and adapted over millions of years.

18. Note on troposphere

The lowest layer of the Earth's atmosphere is called the troposphere. It is where all weather conditions occur and where conventional aviation takes place. The troposphere contains about 75% of the atmosphere's mass and 99% of the total mass of water vapor and aerosols.

Its **height** varies, reaching around 18 kilometers (11 miles) in the tropics, 17 kilometers (11 miles) in the middle latitudes, and 6 kilometers (3.7 miles) in the polar regions during winter. The total average height of the troposphere is 13 km.

Properties:

- In the troposphere, the air composition is mostly uniform except for varying levels of water vapor.
- The temperature generally decreases as you go higher in altitude.
- The ground absorbs most of the sun's energy, warming the lower levels of the atmosphere.
- Heat is also radiated away from the Earth's surface at the top of the atmosphere, causing that part of the atmosphere to cool down.
- The vapor pressure decreases as the temperature drops. This drop is due to gravity.
- The atmospheric pressure is highest at sea level and decreases as you go higher in altitude.

The **tropopause** marks the boundary between the troposphere and the stratosphere. Its temperature decreases with increasing altitude. In equatorial regions, the tropopause is around 17 kilometers (11 miles) above the Earth's surface, while over the polar regions, it is at approximately 9 kilometers (5.6 miles).

19. Carbon Cycle

The carbon cycle is the process by which carbon moves from the atmosphere into living organisms and the Earth, and then back into the atmosphere again.

Carbon is main component of living organisms. 20% of human body is carbon by weight. CO₂ is major source of carbon. Atmosphere contains about 750 billion metric tons of carbon in form of CO₂. Photosynthetic organisms take in CO₂ from the atmosphere

Synthesize the carbon containing organic compounds This process is sometimes called carbon **fixation**. Animals eat plants and make their own organic compounds. Production of energy by breakdown of these organic compounds in respiration.

Respiration releases CO₂ in air. Decaying organisms also produce CO₂.

CO₂ released in air is again available for plants. In aquatic ecosystem, CO₂ reacts with the water to form bicarbonate ions. Dissolved CO₂ and bicarbonates are used by algae and aquatic plants in photosynthesis.

Methanogens, for example, are microbes that produce methane (CH₄) instead of CO₂. Methane released in air is oxidized into CO₂. Amount of CO₂ made by breakdown of organic compounds almost matches the amount of CO₂ used. Small mis-matches can have large consequences.

- Carbon enters the atmosphere as CO₂.
- CO₂ is absorbed by autotrophs such as green plants.
- Animals consume plants, thereby, incorporating carbon into their system.
- Animals and plants die, their bodies decompose and carbon is reabsorbed back into the atmosphere.

20. Detail note on acid rain

Acid rain is a type of rain that contains harmful chemicals. It forms when pollutants, such as sulfur dioxide and nitrogen oxides, are released into the air from activities like burning fossil fuels and industrial processes.

When these pollutants mix with water vapor in the air, they create acids like sulfuric acid and nitric acid. When it rains, these acids come down with the raindrops, making the rain acidic. Acid rain can also fall as snow, sleet, or fog.

Acid rain is caused by both natural and human activities. **Natural causes** include volcanic eruptions releasing sulfur dioxide (SO₂), nitrogen fixation adding nitrogen dioxide (NO₂) to the air, and electrical activity in the atmosphere producing NO₂. Acid deposits in glacial ice can also contribute to acid rain.

Human activities are the main contributors to acid rain. Sulfur and nitrogen compounds from sources like electricity generation, factories, and motor vehicles play a significant role. Burning coal in power plants is the largest source of sulfur and nitrogen compounds. These gases can travel long distances in the atmosphere before being converted to acids and deposited. Tall factory funnels disperse smoke even farther.

Chemical processes, such as gas phase chemistry, also contribute to the formation of acid rain. For example, sulfur dioxide can be oxidized by the hydroxyl radical to form the hydroxysulfonyl radical.

Acid rain can have negative effects on the environment and our surroundings. It can harm plants, trees, and crops by damaging their leaves, stems, and roots. It can also make bodies of water, like lakes and rivers, more acidic, which can harm aquatic life such as fish and amphibians.

Additionally, acid rain can corrode buildings, statues, and other structures made of stone or metal. It can damage the paint on cars and buildings as well.

21. Effects of global warming

Global warming is the long-term increase in Earth's average surface temperature, primarily due to human activities that release greenhouse gases.

Global warming is having significant impacts on our planet. Here are some of the effects of global warming:

1. Rising temperatures: Global temperatures have been steadily rising over the past century, and this trend is expected to continue. This can lead to a range of impacts, including heat waves, droughts, and more frequent and severe wildfires.

2. Melting ice: As temperatures rise, the amount of ice in the Arctic and Antarctic regions is decreasing. This can lead to rising sea levels and changes in ocean currents, which can have significant impacts on weather patterns and ocean ecosystems.

3. Changes in precipitation: Global warming is also causing changes in precipitation patterns, with some areas experiencing more frequent and severe droughts, while others are experiencing more intense rainfall events.

4. More frequent and severe weather events: Climate change is leading to more frequent and severe weather events, such as hurricanes, tornadoes, and floods. These events can cause significant damage to infrastructure and result in loss of life.

5. Impacts on wildlife: As temperatures and precipitation patterns change, wildlife is being impacted as well. Some species are being forced to migrate to new areas in search of food and water, while others are struggling to adapt to changing conditions.

6. Impacts on human health: Global warming can also have significant impacts on human health, particularly in areas that are experiencing extreme heat or changes in precipitation patterns. This can lead to increased rates of heat stroke, respiratory illnesses, and other health problems.

22. Water Pollution:

Any physical or chemical change in water that adversely affects the health of humans and other organisms • Varies in magnitude by location • Global issue • Lack of disease-free water anywhere in the world.

Categories

- 1) Sewage
- 2) Disease-causing agents
- 3) Sediment pollution
- 4) Inorganic plant and algal nutrients
- 5) Organic compounds
- 6) Inorganic chemicals
- 7) Radioactive substances
- 8) Thermal pollution

Causes & Impacts:

Sewage: Sewage contains organic matter, nitrogen, and phosphorus, leading to eutrophication and health risks.

Disease-causing agents: Bacteria, viruses, and parasites from untreated sewage contaminate water bodies and causes illness.

Sediment pollution: Soil and particles from erosion or human activities affect water clarity and aquatic habitats.

Inorganic plant and algal nutrients: Nitrogen and phosphorus from runoff lead to eutrophication and harmful algal blooms.

Organic compounds: Pesticides, herbicides, and industrial chemicals pose toxicity risks to aquatic organisms and human health.

Inorganic chemicals: Heavy metals and acids from industrial discharge and mining are toxic to aquatic organisms and humans.

Radioactive substances: Uranium, radium, and other radioactive materials from natural or human sources pose risks to aquatic life and human health.

Thermal pollution: Increased water temperature due to human activities or natural processes harms aquatic organisms and habitats.

23. Write different types of wastewater treatment? Primary & secondary waste water treatment?

Wastewater Treatment

It is the process of converting wastewater – water that is no longer needed or is no longer suitable for use – into bilge water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

Wastewater treatment is the process of removing contaminants and pollutants from used water to make it safe for discharge back into the environment or for reuse. It involves a combination of physical, biological, and chemical processes to remove solids, organic matter, nutrients, and pathogens, ensuring the protection of public health and the environment.

Processes

- 1) Primary
- 2) Secondary
- 3) Advanced

❖ Primary treatment:

Primary treatment in wastewater treatment involves mechanical processes such as screening, grinding, grit chamber, and sedimentation tank. Screening and grinding remove large debris and break down solids, while the grit chamber settles heavy particles and sedimentation tanks separate suspended solids. Chlorination may be used to disinfect the effluent. Primary treatment reduces pollutant load but is typically followed by secondary and tertiary treatment for further purification.

❖ Secondary treatment:

In secondary treatment using trickling filters, the wastewater from the sedimentation tank is passed through a bed of stones, where bacteria grow and consume organic matter. Cleaner water then trickles through pipes at the bottom for further treatment.

In secondary treatment using activated sludge, the wastewater is pumped into an aeration tank and mixed with air and bacteria-rich sludge called "activated sludge." The bacteria decompose the raw sewage, reducing organic and biological oxygen demand. The mixture, called mixed liquor, then goes to another sedimentation tank where suspended solids settle out for reuse as activated sludge. These processes help further purify the wastewater before its safe release or reuse.

❖ Advanced tertiary treatment:

Advanced tertiary treatment involves using physical and chemical processes to target specific pollutants that remain in wastewater after primary and secondary treatment. This treatment is quite expensive, costing twice as much as building a secondary treatment plant. The processes used in advanced tertiary treatment include bleaching to remove coloration, disinfection to kill pathogens, coagulation-sedimentation with alum, adsorption using activated charcoal, and electrodialysis for salt removal. These processes help ensure that the wastewater is thoroughly treated before it is released back into the environment.

24. What are parasites?

Parasitism is a type of symbiotic relationship between two species in which one species, the parasite, benefits at the expense of the other species, the host.

categories

Micro and macro are two categories of parasites based on their size and complexity.

- 1) **Micro parasites** are small and simple parasites that can only be seen with a microscope. They include things like bacteria, viruses, and protozoa. Micro parasites can cause a wide range of diseases in their hosts, such as malaria, Lyme disease, and the common cold.
- 2) **Macro parasites**, on the other hand, are larger and more complex parasites that can be seen with the naked eye. They include things like worms, ticks, and lice. Macro parasites can also

cause a variety of health problems in their hosts, such as skin irritation, anemia, and malnutrition.

Types

There are different types of parasites based on where they live and how they affect their hosts. Here are some examples:

- 1) **Ectoparasites:** These are parasites that live on the surface of the host's body, such as fleas, ticks, and lice.
- 2) **Endoparasites:** These are parasites that live inside the host's body, such as tapeworms, roundworms, and hookworms.
- 3) **Obligate parasites:** These are parasites that cannot survive without a host, such as lice and tapeworms.
- 4) **Facultative parasites:** These are parasites that can live either on a host or in the environment, such as some fungi and bacteria.
- 5) **Social parasites:** These are parasites that exploit the social behavior of their host, such as cuckoo birds that lay their eggs in the nests of other birds.

25. Aposematism

Aposematism is a defensive strategy or appearance of an animal that warns predators it is toxic, distasteful or dangerous.

Aposematism is the advertising by an animal to potential predators that it is not worth attacking or eating. This unprofitability may consist of any defences which make the prey difficult to kill and eat, such as toxicity, venom, foul taste or smell, sharp spines, or aggressive nature.

Role:

- Function is to prevent attack.
- Beneficial for both predator and prey, saving them from harm.

Advertisement of signals:

- 1) **Coloration:** Many animals use bright colors to signal that they are dangerous or unpalatable. For example, poison dart frogs have bright, contrasting colors to warn predators that they are toxic.
- 2) **Odor:** Some animals use strong odors to signal their unpalatability. For example, skunks release a strong-smelling spray when threatened.
- 3) **Sound:** Some animals use loud or distinctive sounds to warn predators that they are dangerous. For example, rattlesnakes use their rattles to warn predators to stay away.
- 4) **Behavior(Aggression):** Some animals use specific behaviors to signal their unpalatability. For example, the death feigning behavior of some insects signals that they are dead or dying, which can deter predators.

• Types of Signals

Aposematic signals can be classified as either primary or secondary signals.

- 1) **Primary:** Easily detectable (colors)
Primary signals are those that are directly associated with the unpalatable or dangerous qualities of the animal, such as bright colors, strong odors, or toxic secretions and distinctive patterns to attract attention and communicate danger.
- 2) **Secondary:** invisible (ultrasonic noise for specific predator)
Secondary signals, on the other hand, are those that are associated with the primary signals, but are not directly related to the unpalatability or danger of the animal. For example, some animals may use bright colors to signal their unpalatability, but may also use other secondary signals, such as distinctive body shapes or patterns, to enhance the effectiveness of their warning.

26. Give the details of invasive species

Invasive species are plants, animals, or other organisms that are not native to a particular ecosystem but are introduced to it, often by human activities. These species can spread quickly and have negative impacts on the environment, economy, and even human health.

Invasive species can be introduced to a new environment through various means, such as accidental or intentional release, escape from captivity, or as hitchhikers on ships, planes, or vehicles. Once they are introduced, they can spread quickly and easily because they have no natural predators or diseases to control them. They can also adapt to new environments and outcompete native species for resources such as food, water, and space.

Invasive species can cause harm to the environment by altering habitats and changing the way ecosystems work. For example, invasive plants can grow quickly and block sunlight from reaching the ground, which can prevent native plants from growing. Invasive animals can prey on native species or compete with them for food, which can lead to the decline or extinction of native species. Invasive species can also spread diseases to native species, which can cause further damage to the ecosystem.

Invasive species can also have economic impacts. For example, invasive insects can damage crops, invasive plants can reduce timber production, and invasive animals can reduce fish populations. These impacts can cause financial losses for industries such as agriculture, forestry, and fisheries..

27. Feeding level of food chain

Food energy passes through different sequence.

Example: plants → grass hopper → frog → snake → hawk

Many plant species eaten by insects. Many insects species eaten by frogs

Food Chain

Linear network of links to pass energy in form of food. Simplified abstractions of real food webs, but complex in their dynamics

Feeding Levels

This complexity is organized by ecologists. Recognized limited number of feeding or trophic levels

Food Chain

Food energy passes through different sequence.

- Level 1: Primary producer
- Level 2: Primary consumer
- Level 3: Secondary consumer
- Level 4: Tertiary Consumer
- Level 5: Apex Predator

A food chain also shows how the organisms are related with each other by the food they eat. Each level of a food chain represents a different trophic level. Food chain only follows a direct, linear pathway of one animal at a time.

Detritivore and decomposers can be at trophic level as well. They return nutrients back in soil to be available for producers.

28. Rates of Evolution?

• Evolutionary changes are occurring gradually over millions of years. • This concept, called phyletic gradualism. • It has been the traditional interpretation of the tempo, or rate, of evolution. • Species do not change significantly over millions of years. • Stasis is interrupted by ecological crisis, such as a change in climate or a major geological event. • At this interruption, variation that previously was selectively neutral or disadvantageous might now be advantageous. • Geological events might result in new habitats becoming available. • Brief period of interruption punctuates long periods of stasis. • It is called as punctuated equilibrium model of evolution. • Stabilizing selection characterizes the periods of stasis. • Directional or disruptive selection characterizes the periods of change. • Rapid evolutionary changes in small populations. • Example: rapid pesticide and antibiotic resistance • Galapagos islands finches • A long, dry period from the middle of 1976 to early January 1978 resulted in birds with larger, deeper bills. • Over one year time period • Initially, birds consumed smaller seeds. • After depletion of small seeds, they had to eat large seeds. • Small beaks developed into large beaks. • This period of rapid change is followed by stasis again. • It explains missing transitional stages in fossil record.

29. Explain Interactions Among Evolutionary Forces.

• The amount of genetic variation in a population may be determined by the relative strength of different evolutionary processes. • Sometimes these processes act together, and in other cases they work in opposition.

❖ Mutation and genetic drift may counter selection

• Both genetic drift and natural selection remove alleles from population. • Natural selection is non random • It operates to increase the representation of alleles that enhance survival and reproductive success. • Genetic drift is random. • In which any allele may increase or depart the population. • Sometimes, drift may lead to a decrease in the frequency of an allele that is favored by selection. • Mutation may also counter natural selection. • It may randomly decrease or increase the frequency of alleles.

❖ Gene flow may promote or constrain evolutionary change

• Gene flow can be either a constructive or a constraining force. • It can spread a beneficial mutation that arises in one population to other populations. • Gene flow can impede adaptation within a population by the continual flow of inferior alleles from other populations. • Natural selection is removing inferior alleles while gene flow can bring them back. **Example:** • Heavy metals tolerance

30. Which limiting factor effects population size?

• Factors limit the population size and growth. 1. Density dependent factors 2. Density independent factors

❖ **Density dependent factors**
• Example: Food

Competition within the population • When a population reaches a high density, there are more individuals trying to use the same quantity of resources. • This can lead to competition for food, water, shelter, mates, light, and other resources needed for survival and reproduction.

Predation • Higher-density populations may attract predators who wouldn't bother with a sparser population. • When these predators eat individuals from the population, they decrease its numbers but may increase their own.

Disease and Parasites • Disease is more likely to break out and result in deaths when more individuals are living together in the same place.

Waste Accumulation • High population densities can lead to the accumulation of harmful waste products that kill individuals or impair reproduction, reducing the population's growth.

❖ **Density Independent Limiting Factors**

Affect per capita growth rate independent of density of population • Natural disasters • Severe weather • Pollution . Cannot regulate population at constant level . They are accidental. Cause abrupt shift in population size. May wipe out small population by sporadic event • Collectively, densitydependant and independent factors create a different population growth curve. • Cyclic oscillations

31. Explain logistic growth population

Logistic population growth is influenced by the carrying capacity of the environment, which determines the maximum population size an ecosystem can sustain. Environmental resistance, including factors like limited resources, competition, predation, and disease, slows down population growth as it approaches the carrying capacity. As the population nears its limit, the growth rate decreases due to the constraints imposed by environmental factors. Various elements, such as climate, availability of food, space, and competition for resources, impact the population growth. These factors interact to shape the dynamics of population size and distribution in a given ecosystem.

The formula for Logistic growth is represented by $dN/dt = rN (K-N)/K$

where N is the population size, r is the intrinsic rate of natural increase, K is carrying capacity and dN/dt is change in population size per unit time.

32. Explain exponential growth.

Exponential growth occurs when resources are abundant, allowing a population to increase rapidly and continuously. During exponential growth, the population size experiences a remarkable and unrestricted increase over time. However, this growth is not sustainable indefinitely and is ultimately limited by the availability of resources.

The formula for exponential growth is represented by $dN/dt = rN$

where N is the population size, r is the intrinsic rate of natural increase, and dt represents the change in population size over time. This formula quantifies the exponential addition to the population as it grows.

EXPONENTIAL GROWTH	LOGISTIC GROWTH
It occurs when resources are abundant.	It occurs when resources are limited.
Population passes well beyond the carrying capacity of the ecosystem.	Population doesn't grow beyond the carrying capacity of the ecosystem.
It produce J shaped curve.	It produce S shaped curve.
A stationary or stable phase is seldom achieved.	A stationary or stable phase is achieved.
It has two phases lag and log.	It has four phases lag, log, deceleration & steady .
It occurs in fewer organisms like lemmings, algal blooms, etc.	It is more common, example members of wildlife, yeast (a microscopic fungus), etc.